MIL-C-24231/5D(SH)
15 May 1987
SUPERSEDING
MIL-C-24231/5C(SHIPS)
30 August 1976
(See note 4)

MILITARY SPECIFICATION SHEET

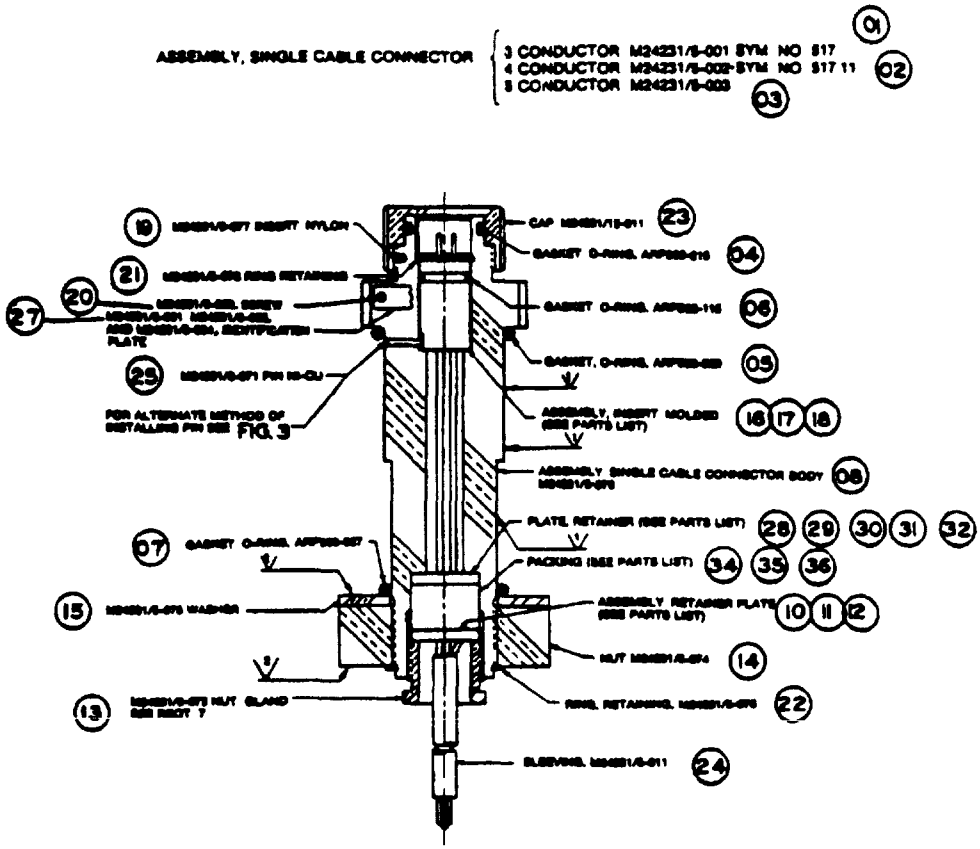
CONNECTORS, SINGLE CABLE, TYPE III, THREE-, FOUR- OR FIVE-CONDUCTOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the single cable connector described herein shall consist of this specification and the latest issue of MIL-C-24231.

AMSC N/A FSC 5935
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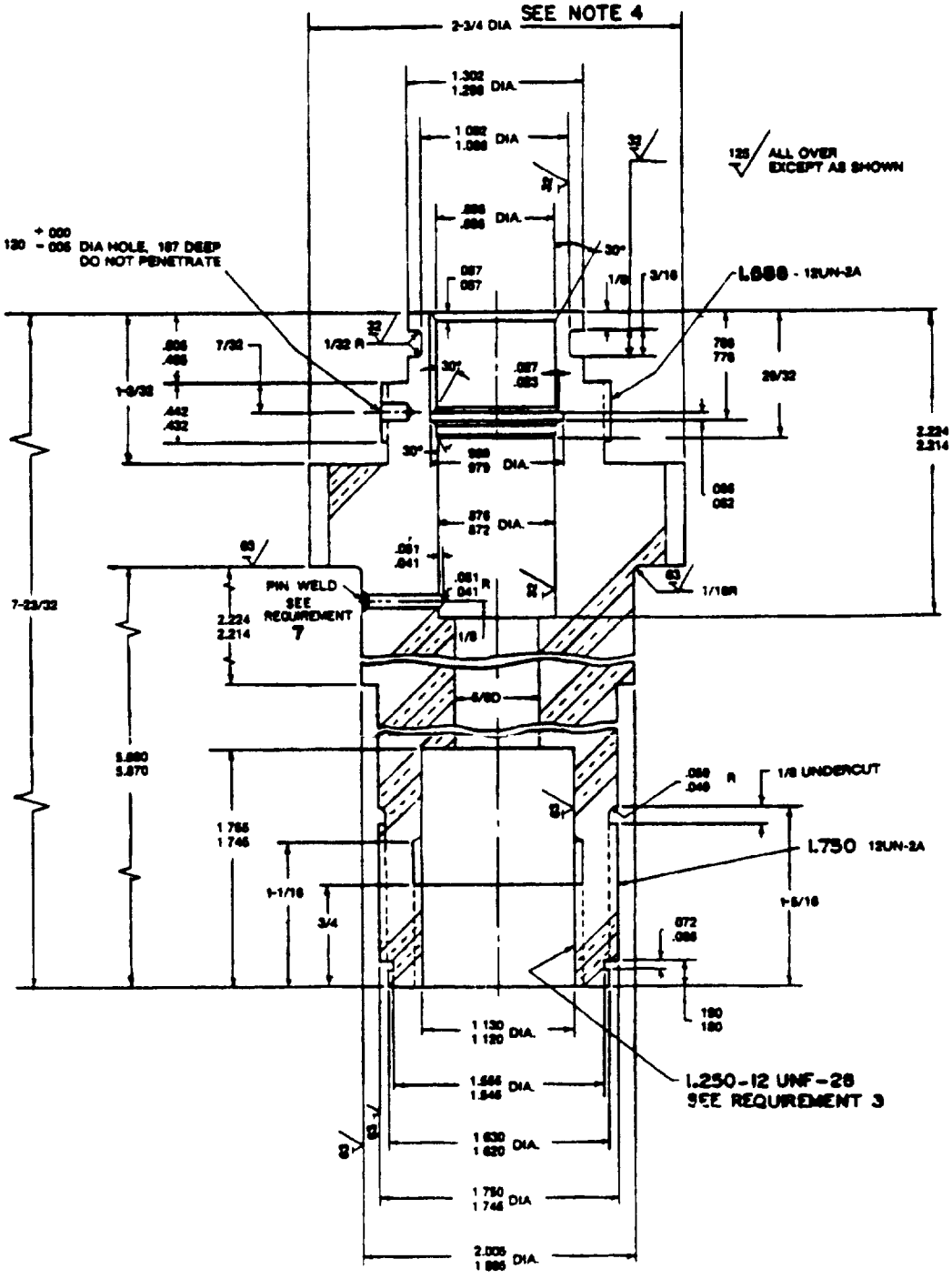
\1/ Hardness reading to be taken at these points.

\2/ Hardness reading to be taken at this face at 120-degree intervals.

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FIGURE 1. Connector, single cable - typical.

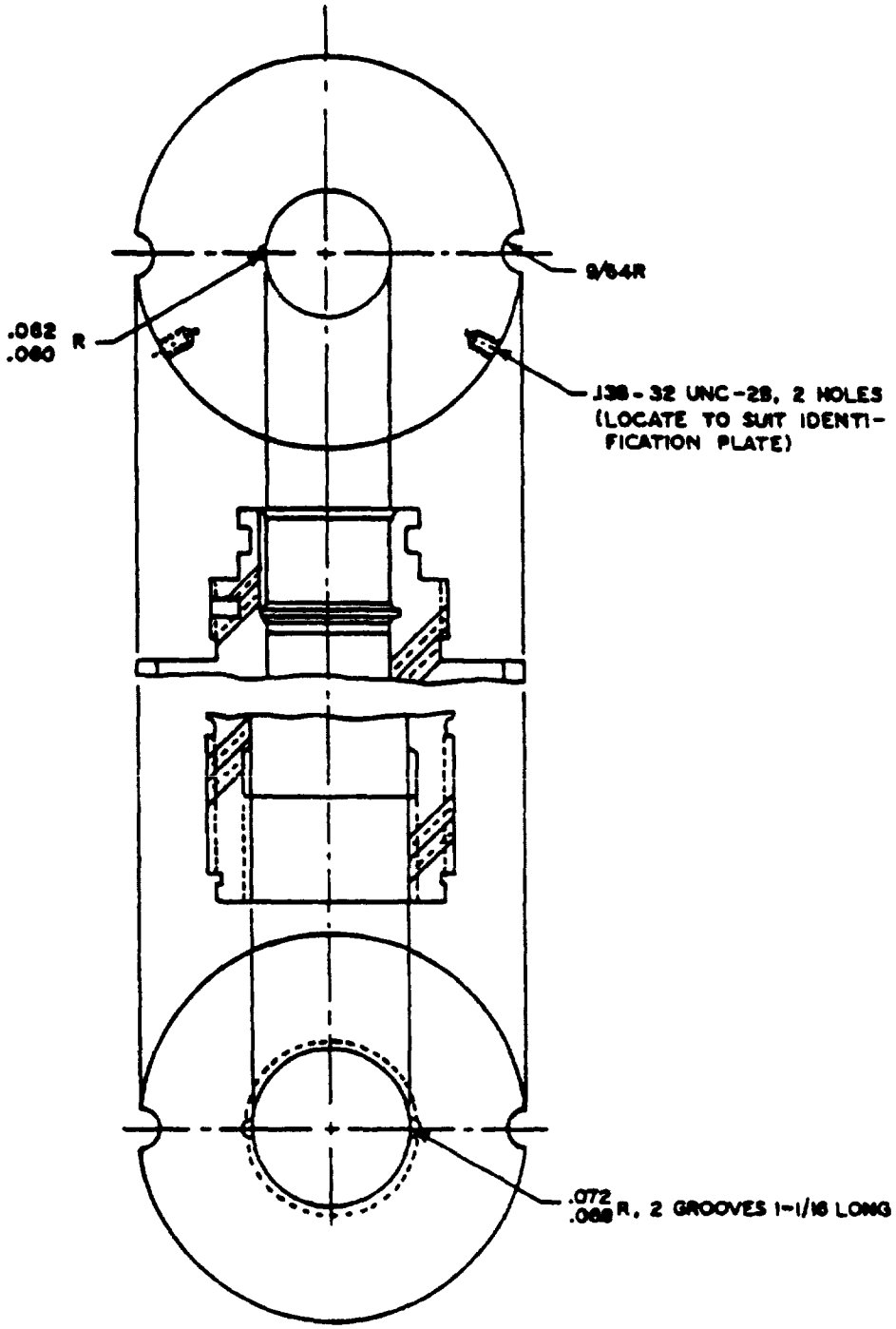
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FIGURE 2. Assembly, single cable connector body - M24231/5-070. (08)

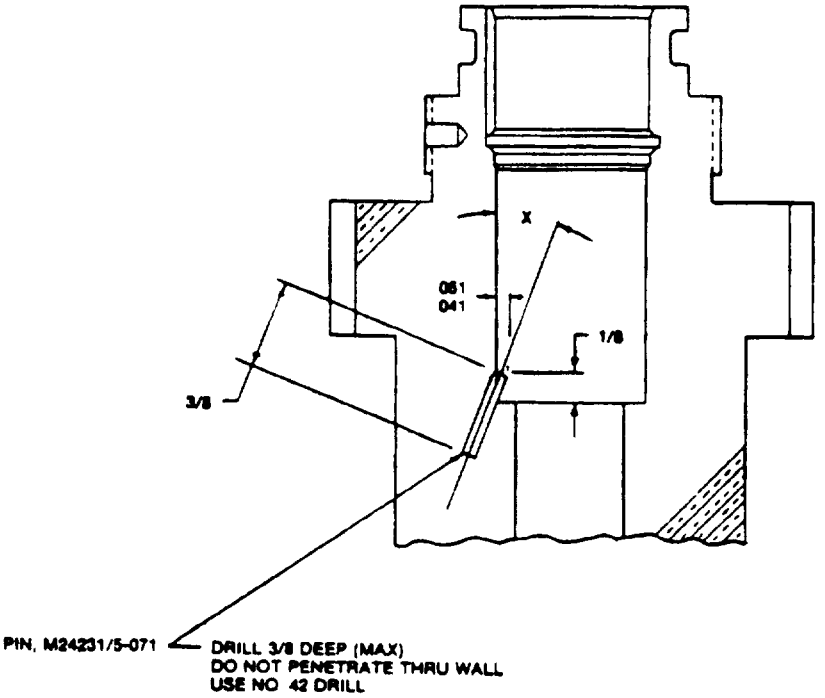
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FIGURE 2. Assembly, single cable connector body - M24231/5-070. - Continued (08)

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CONNECTOR BODY	"X" DEGREES
M24231/5-070	21 1/2
M24231/6-010	21 1/2
M24231/7-040	27
M24231/8-020	27
M24231/9-030	27
M24231/9-040	27
M24231/10-040	35
M24231/19-010	40

NOTE: Install pin with locking and retaining compound in accordance with grade A of MIL-S-22473.

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FIGURE 3. Alternate method for installation of guide pin (see requirement 4). (25)

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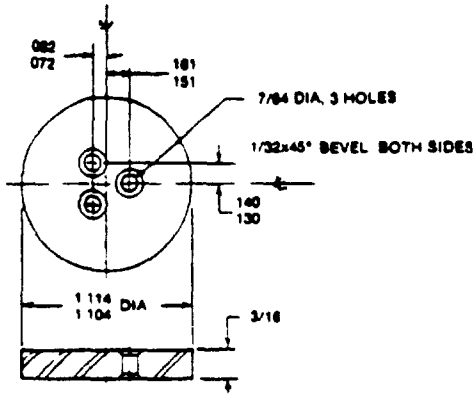


FIGURE 4. Plate, retainer - M24231/5-051. (28)

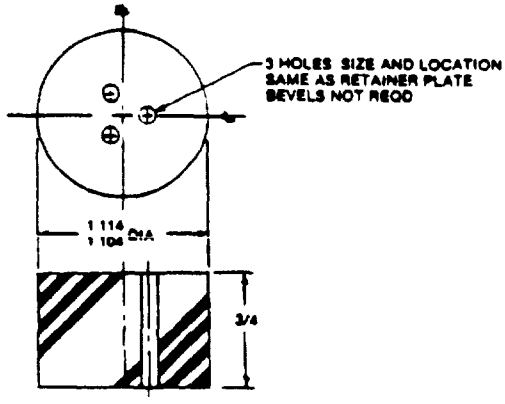
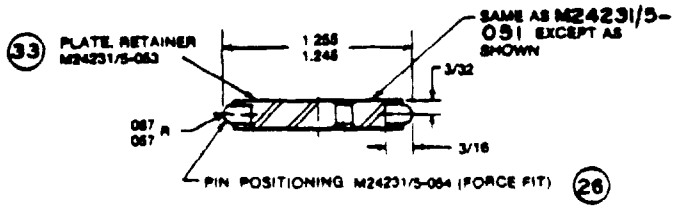


FIGURE 5. Packing - M24231/5-052. (34)



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FIGURE 6. Assembly, retainer plate - M24231/5-050. (10)

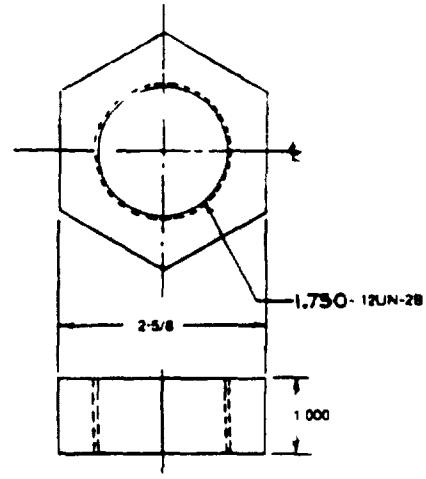


FIGURE 7. Nut - M24231/5-074. (14)

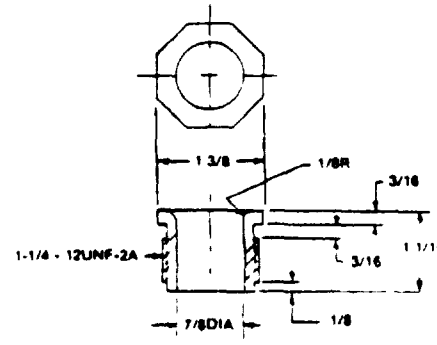


FIGURE 8. Nut, gland - M24231/5-072. (13)

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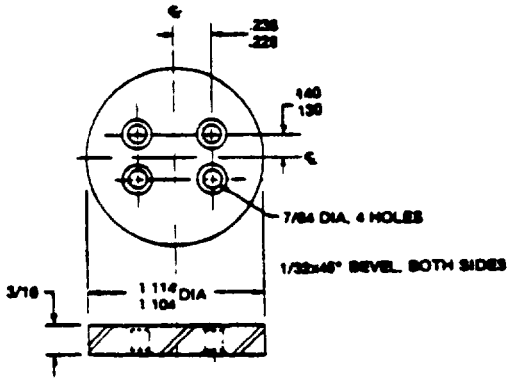


FIGURE 9. Plate, retainer - M24231/5-061. (29)

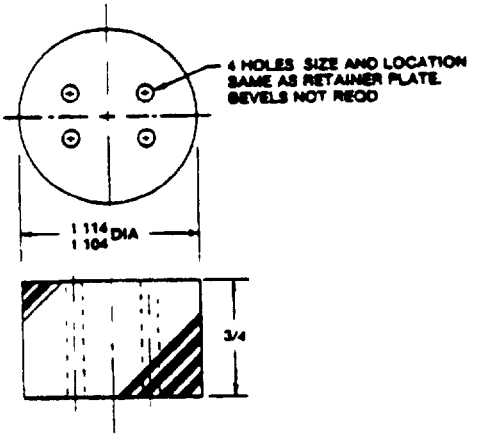
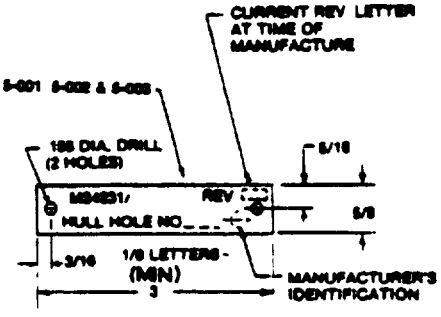
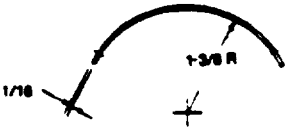


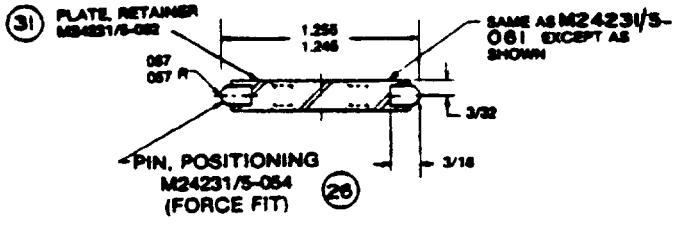
FIGURE 10. Packing - M24231/5-063. (35)





M24231/5-061 (FOR M24231/5-001)
M24231/5-062 (FOR M24231/5-002)
M24231/5-064 (FOR M24231/5-003)

FIGURE 12. Plate, identification. (27)



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FIGURE 11. Assembly, retainer plate - M24231/5-060. (11)

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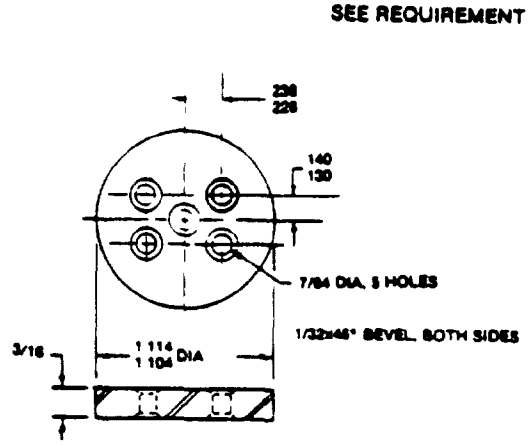


FIGURE 13. Plate, retainer - M24231/5-064. (30)

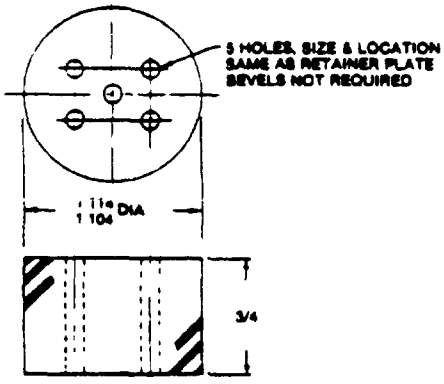
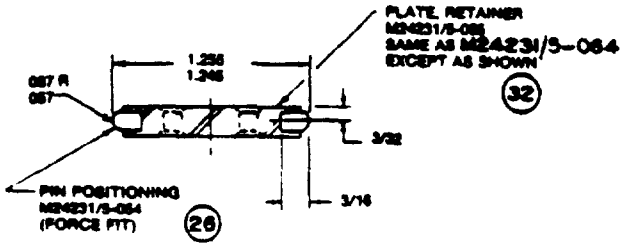
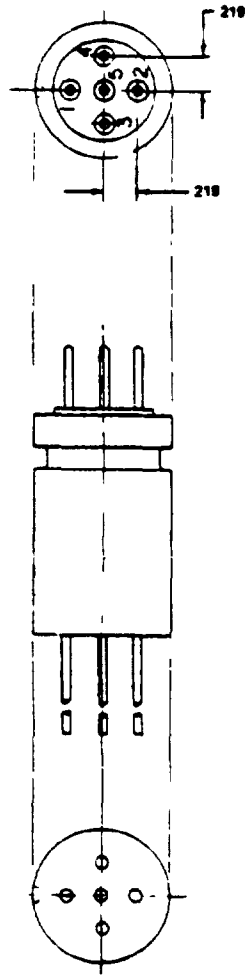


FIGURE 14. Packing - M24231/5-085. (36)



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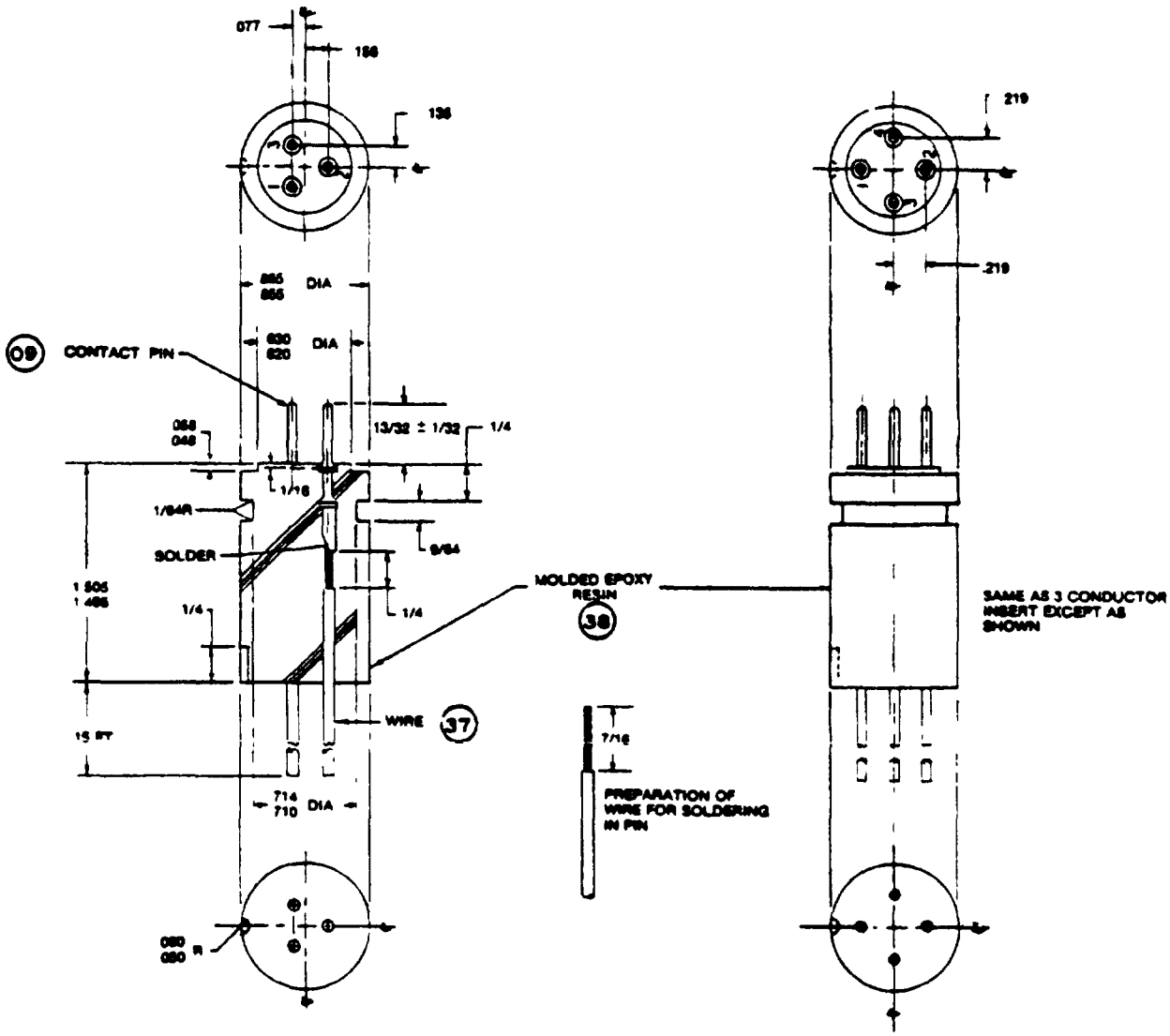
FIGURE 15. Assembly, retainer plate - M24231/5-066. (12)



(18)

FIGURE 16. Assembly, insert molded (5-conductor) - M24231/5-067.

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FIGURE 17. Assembly, insert, molded (3-conductor) - M24231/5-010. (16)

FIGURE 18. Assembly, insert, molded (4-conductor) - M24231/5-020. (17)

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REQUIREMENTS:

1. Dimensions and configuration: See figures.
2. The gland nut shall be tightened until the packing tightens against wires. This can be determined by moving wires gently back and forth until wire is seized by packing. The gland nut shall be turned approximately one-fourth of a turn more to hold the wires in place. Do not overtighten the gland nut or attempt to tighten down metal to metal, as this may damage the wires.
3. Full or partial threads shall not extend more than 0.063 inch beyond the 3/4-inch thread length.
4. The alternate method of installing the guide pin (see figure 3) eliminates the need for welding in the fabrication of the single cable connector body.
5. Other special 5-conductor arrangements (for example, 2SWU-1 and B-16 wire combinations) shall be the responsibility of the installing activity.
6. The retainer plates and retainer plate assemblies shall be manufactured from nickel-copper alloy in accordance with class A, form 1, hot finished condition of QQ-N-281 with a minimum yield strength of 40,000 lbf/in^2 at 0.2 percent offset, or nickel-copper alloy in accordance with class A, form 1, cold drawn condition of QQ-N-281 with a minimum yield strength of 50,000 lbf/in^2 at 0.2 percent offset.
7. Welding of the pin (/5-071) is allowed after the hull fitting body has been age-hardened to achieve its required physical properties.

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TABLE I. Parts list.

Item 1/ no.	Description 2/	Quantity for M24231/5- 3/ 001	002	003	Material/specification	Remarks
01	Connector assembly, single cable, 3-conductor, symbol no. 517	1			-----	See figure 1
02	Connector assembly, single cable, 4-conductor, symbol no. 517-11		1		-----	See figure 1
03	Connector assembly, single cable, 5-conductor			1	-----	See figure 1
04	O-ring gasket	1	1	1	MS28775-215	Lubricate with silicone compound, MIL-S-8660, see figure 1
05	O-ring gasket	1	1	1	MS28775-329	See item 04
06	O-ring gasket	1	1	1	MS28775-115	See item 04
07	O-ring gasket	1	1	1	MS28775-327	See item 04
08	Body, single cable connector	1	1	1	Ni-Cu-Al, Class A, QQ-N-286	See figure 2
09	Pin, contact	3	4	5	MIL-C-24231	See figure 17
10	Assembly, plate, retainer	1			See items 26 and 33	See figure 6 and requirement 6
11	Assembly, plate, retainer		1		See items 26 and 31	See figure 11 and requirement 6
12	Assembly, plate, retainer			1	See items 26 and 32	See figure 15 and requirement 6
13	Nut, gland	1	1	1	Ni-Al-BRZ, QQ-C-465	See figure 8 and requirement 2

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/5- 3/ 001	002	003	Material/specification	Remarks
14	Nut, body	1	1	1	Ni-Cu-Al, class A, QQ-N-286	See figure 7
15	Washer, body, 3.5 od x 1.760 id by 0.162 thick	1	1	1	Ni-Cu-Al, class A, QQ-N-286	See figure 1
16	Insert assembly molded	1			See items 09, 37, and 38	See figure 17
17	Insert assembly molded		1		See items 09, 37, and 38	See figure 18
18	Insert assembly molded			1	See items 09, 37, and 38	See figure 16
19	Insert, 0.125 ± 0.002D, Lg to suit	1	1	1	Nylon, group I, class I, ASTM D 4066	See figure 1
20	Screw, 0.138-32x3/16 Lg RHMS	2	2	2	CRES, coml	See figure 1
21	Ring, retaining	1	1	1	SPIRALOX, PRD-99-M, or equal	See figure 1, M24231/5-073
22	Ring, retaining	1	1	1	TRUARC, type 5100-162-W, or equal	See figure 1, M24231/5-076
23	Cap, pressure-proof	1	1	1	M24231/12-011	See figure 1
24	Sleeving, 0.360 id by 0.020 thick, 14 feet	3	4	5	Type F, form U, category 1, color clear, MIL-I-631	See figure 1
25	Pin, guide, 0.092 diameter, Lg to suit	1	1	1	Ni-Cu, class A, QQ-N-281	See figures 1 and 3, requirements 4 and 7
26	Pin, positioning	1	1	1	Ni-Cu, class A, QQ-N-281	See figures 6, 11, and 15

See footnotes at end of table.

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TABLE I. Parts list. - Continued

Item 1/ no.	Description 2/	Quantity for M24231/5- 3/ 001	002	003	Material/specification	Remarks
27	Plate, identification 1/16 thick	1	1	1	CRES or Al, type I, MIL-P-15024	See figure 12
28	Plate, retainer	1			Ni-Cu, class A, QQ-N-281	See figure 4 and requirement 6
29	Plate, retainer		1		Ni-Cu, class A, QQ-N-281	See figure 9 and requirement 6
30	Plate, retainer			1	Ni-Cu, class A, QQ-N-281	See figure 13 and requirement 6
31	Plate, retainer		1		Ni-Cu, class A, QQ-N-281	See figure 11 and requirement 6
32	Plate, retainer			1	Ni-Cu, class A, QQ-N-281	See figure 15 and requirement 6
33	Plate, retainer	1			Ni-Cu, class A, QQ-N-281	See figure 6 and requirement 6
34	Packing, durometer, 55 ± 5	1			Rubber, ZZ-R-768	See figure 5
35	Packing, durometer, 55 ± 5		1		Rubber, ZZ-R-768	See figure 10
36	Packing, durometer, 55 ± 5			1	Rubber, ZZ-R-768	See figure 14
37	Wire, 15 feet, 1 inch	3	4	5	Type B-16, MIL-W-16878, nylon jacket, black	See figure 17
38	Epoxy resin, molded	As required			Cast epoxy resin comp 4/	See figures 16, 17, and 18

1/ Item numbers are the same as encircled numbers shown on figures.
2/ See applicable figure or table listed in remarks column for part numbers and dimensions.
3/ Quantity is for one complete assembly of each size.
4/ Arcoset No. 200, M.M. Reif and Co., Rockledge, PA, or Phillicast No. 2, Philadelphia Resin Corp., Montgomeryville, PA, or equal.

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NOTES:

1. All dimensions are in inches.
2. Unless otherwise indicated, tolerances shall be as follows:

   (a) Decimals - plus or minus 0.005 inch.
   (b) Fractions - plus or minus 1/64 inch.
   (c) Angular dimensions - plus or minus 1/2 degree.

3. The flange outside diameter is intended to coincide with commonly available 2-3/4 inch bar stock. The dimension may vary after machine clean-up.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 5935-N279-05)

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